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Specialized Topics of Theoretical Physics

Majorana Modes in the Machine –
Simulating Topological Phases with Quantum Circuits and AI

Finite-Size Physics & Majorana Edge Modes

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1 Introduction

The 1D Kitaev chain is a toy model of a topological superconducting wire, showing how Majorana zero modes arise at its ends. **Part 1** studied the infinite chain with periodic boundaries, where the phase $|\mu| < 2t$ is fixed by a bulk invariant. A real wire is finite and open, and that finiteness is what turns ideal zero modes into something measurable: this report treats the spectrum of the finite open chain, the edge modes and their exponential localisation, the hybridisation splitting $E_0(L)$, and the sign-filter artefact that appears once the splitting drops below machine precision. Beyond the usual sweet-spot ($t = \Delta$) treatment, the analytic backbone is the exact finite-L theory of Leumer *et al.* [4, 5]: a determinant criterion locating the L discrete "Majorana lines" on which the splitting vanishes *exactly*, and a quantization rule treating bulk and edge states on the same footing.

2 Theory: edge modes of the finite open chain

2.1 BdG form of the finite chain

We consider the Kitaev chain on L sites with open boundary conditions,

$$H_K = -\mu \sum_{j=1}^{L} c_j^\dagger c_j - t \sum_{j=1}^{L-1} \big(c_j^\dagger c_{j+1} + \mathrm{h.c.}\big) + \Delta \sum_{j=1}^{L-1} \big(c_j^\dagger c_{j+1}^\dagger + \mathrm{h.c.}\big), \tag{2.1}$$

with real t, Δ and the symmetric ("sweet") point $t = \Delta$. In the Nambu basis $\Psi = (c_1, \dots, c_L, c_1^\dagger, \dots, c_L^\dagger)^T$ it becomes a $2L \times 2L$ quadratic form,

$$H_K = \tfrac{1}{2}\Psi^\dagger M \Psi, \quad M = \begin{pmatrix} h & d \\ -d & -h \end{pmatrix}, \quad h_{ij} = -\mu\,\delta_{ij} - t\,(\delta_{i,j+1} + \delta_{i+1,j}), \quad d_{ij} = \Delta\,(\delta_{j,i+1} - \delta_{i,j+1}), \tag{2.2}$$

with h real symmetric tridiagonal and d real antisymmetric, carrying $+\Delta$ on its upper diagonal so that $\frac{1}{2}\Psi^\dagger M \Psi$ reproduces the pairing term of (2.1). The particle–hole operator $\mathcal{C} = \tau_x \mathcal{K}$ (τ_x acting on the particle–hole blocks, $\mathcal{K}$ complex conjugation) gives $\mathcal{C} M \mathcal{C}^{-1} = -M$, so $M\psi = E\psi$ implies $M(\mathcal{C}\psi) = -E(\mathcal{C}\psi)$: the $2L$ eigenvalues come in $\pm E$ pairs, each pair *one* physical quasiparticle rather than two, so each pair must be counted once.

2.2 Edge mode and localisation length at the sweet spot

The Ansatz $\psi_j \sim \lambda^j$ reduces the bulk BdG equations to a 2×2 block with $A = -\mu - t(\lambda + \lambda^{-1})$ on the diagonal and $B = -\Delta(\lambda - \lambda^{-1})$ off it; $B^2 - A^2 = 0$ is the characteristic equation

$$\big[-\mu - t(\lambda + \lambda^{-1})\big]^2 - \Delta^2(\lambda - \lambda^{-1})^2 = 0. \tag{2.3}$$

At $t = \Delta$ this factorises as $(A - B)(A + B) = 0$, giving the branches $\lambda = -\mu/2t$ and $\lambda = -2t/\mu$; the left edge mode is the decaying one, $|\lambda| < 1$:

$$\psi_j^L \propto \lambda^{j-1}, \qquad \lambda = -\frac{\mu}{2t}, \qquad \xi = \frac{1}{\ln(2t/|\mu|)}, \tag{2.4}$$

which exists precisely inside the topological phase $|\mu| < 2t$; ξ is the coherence length and $\psi_j^R \propto \lambda^{L-j}$ the mirrored right edge mode. The decay rate is $|\lambda| = |\mu|/2t$; the only structure on the exponential is the sign of λ, a $(-1)^j$ staggering present for $\mu > 0$ and absent for the $\mu < 0$ used throughout. For $t \neq \Delta$ the roots of (2.3) can be genuinely complex and the splitting oscillates around the envelope [1, 3].

2.3 Hybridisation splitting of the zero mode

In the infinite chain ψ^L and ψ^R are degenerate at exactly $E = 0$; in a finite chain they overlap and the degeneracy is lifted by $\delta E \approx |\langle\psi^L|H|\psi^R\rangle|/(\|\psi^L\|\,\|\psi^R\|)$. The matrix element is boundary-dominated: both modes annihilate the infinite-chain Hamiltonian, and the open chain differs from it only by the bonds removed at the ends, so $H|\psi^R\rangle$ is non-zero only where the right mode has leaked to the left edge, of size λ^{L-1}. The normalisation $\|\psi^L\|^2 \to 1/(1-\lambda^2)$ only sets the prefactor, fixed by the exact finite-chain calculation at $t = \Delta$:

$$\boxed{E_0(L) \approx 2t\left(1-\lambda^2\right)\left(\frac{|\mu|}{2t}\right)^L, \qquad |\lambda| = \frac{|\mu|}{2t}.} \tag{2.5}$$

Three predictions follow: $E_0 = 0$ exactly at $\mu = 0$ for any L; exponential decay in the interior with semilog slope $\ln|\lambda| = -1/\xi$; and, as $|\mu| \to 2t$, $|\lambda| \to 1$ and $\xi \to \infty$, so (2.5) — which assumes $L \gg \xi$ — breaks down and the splitting stops being exponentially small, saturating instead at the finite-size gap $\simeq \pi t/L$. The zero mode is therefore never protected near the transition. The true eigenstates are $(\psi^L \pm \psi^R)/\sqrt{2}$ at $\pm E_0$: the two edge Majoranas jointly form a single non-local fermion, and $E_0(L)$ measures how far a finite system is from hosting true zero modes.

2.4 Exact zero-energy lines for $t \neq \Delta$

Equation (2.5) is asymptotic ($L \gg \xi$) and tied to the sweet spot. The sharper question — *when is E_0 exactly zero at finite L?* — has a closed answer for all t, Δ, μ, due to Leumer *et al.* [4, 5]. Split each fermion into Majorana operators $c_j = (\gamma_j^A + i\gamma_j^B)/2$ and order them by species, $\psi_c = (\gamma_1^A, \ldots, \gamma_L^A, \gamma_1^B, \ldots, \gamma_L^B)^T$. Since (2.1) contains no $\gamma^A\gamma^A$ or $\gamma^B\gamma^B$ terms, the BdG matrix in this *chiral basis* is block *off*-diagonal, $H_K = \frac{1}{4}\psi_c^\dagger H_c \psi_c$ with

$$H_c = \begin{pmatrix} 0 & h_c \\ h_c^\dagger & 0 \end{pmatrix}, \qquad (h_c)_{ij} = -i\mu\,\delta_{ij} - b\,\delta_{j,i+1} + a\,\delta_{i,j+1}, \qquad a = i(\Delta - t), \quad b = i(\Delta + t), \tag{2.6}$$

tridiagonal but *not* Hermitian. (Ref. [5] uses the opposite pairing sign, which transposes h_c; determinants, and hence everything below, are unchanged.) Zero energy is then detected by a determinant rather than the full spectrum: the block structure gives $\det H_c = \det(-h_c h_c^\dagger) = (-1)^L|\det h_c|^2$, so $E = 0$ occurs *iff* $\det h_c = 0$. The leading principal minors of a tridiagonal matrix obey $D_j = -i\mu\,D_{j-1} + ab\,D_{j-2}$ with $ab = t^2 - \Delta^2$, and the Ansatz $D_j \propto (ir)^j$ turns this into $r^2 + \mu r + (t^2 - \Delta^2) = 0$, hence

$$\det h_c = i^L\,\frac{r_+^{L+1} - r_-^{L+1}}{r_+ - r_-}, \qquad r_\pm = \tfrac{1}{2}\Big(-\mu \pm \sqrt{\mu^2 + 4(\Delta^2 - t^2)}\Big), \tag{2.7}$$

which vanishes iff $r_+^{L+1} = r_-^{L+1}$ (for the degenerate case $r_+ = r_-$ the limit is $\det h_c = i^L(L+1)r_+^L$, zero only at $r_+ = 0$). Using $r_+ r_- = t^2 - \Delta^2$ and parametrising $\mu = 2\sqrt{t^2-\Delta^2}\cos\theta$ gives $r_\pm = -\sqrt{t^2-\Delta^2}\,e^{\mp i\theta}$, so the condition reduces to $\sin[(L+1)\theta] = 0$ with $\sin\theta \neq 0$, i.e. $\theta_n = n\pi/(L+1)$:

$$\boxed{\mu_n = 2\sqrt{t^2 - \Delta^2}\;\cos\Big(\frac{n\pi}{L+1}\Big), \qquad n = 1, \ldots, L, \qquad t^2 \geq \Delta^2.} \tag{2.8}$$

A finite open chain therefore carries *exact* zero modes only on L discrete "Majorana lines" [4], and only for $|t| \geq |\Delta|$ — the sole exception being $\mu = 0$ for odd L, where $n = (L+1)/2$ kills the cosine for any t, Δ. Elsewhere in the topological phase E_0 is small but strictly finite. Two consequences matter. At the sweet spot the square root collapses and all L lines merge into $\mu = 0$: there $\{r_+, r_-\} = \{0, -\mu\}$ and $|\det H_c| = \mu^{2L}$, vanishing only at $\mu = 0$ — hence a smooth $E_0(\mu)$ with a single exact zero. For

$|\Delta| < |t|$, instead, E_0 oscillates beneath its envelope and touches zero L times, though only inside the sub-interval $|\mu| \le 2\sqrt{t^2-\Delta^2}$ of the topological window; each crossing is a fermion-parity switch of the ground state [1, 3].

The mode on a line is exact too: $h_c^\dagger \vec{v}_A = 0$ reads $b\,\xi_{j-1} + i\mu\,\xi_j - a\,\xi_{j+1} = 0$, a Fibonacci recursion whose solution with $\xi_0 = 0$ and $\mu = \mu_n$ is [5]

$$\xi_j \propto (-1)^{j-1}\,\frac{\sin(j\theta_n)}{\sin\theta_n}\,e^{-(j-1)/\xi_{\mathrm{loc}}}, \qquad \xi_{\mathrm{loc}} = 2\Big/\ln\left|\tfrac{t+\Delta}{t-\Delta}\right|, \tag{2.9}$$

a pure-γ^A Majorana bound to one edge, its γ^B partner ($a \leftrightarrow b$, $\mu \to -\mu$) decaying from the other. The envelope is exponential, but the chemical potential survives inside $\sin(j\theta_n)$ as a real-space *oscillation* of the wavefunction — the lattice analogue of the factor $e^{ik_0 j}$ absent in (2.4).

2.5 The finite-size quantization rule

The determinant settles only the $E = 0$ question. The full finite-L spectrum follows from squaring the problem to $h_c h_c^\dagger \vec{v} = E^2 \vec{v}$, whose amplitudes need a four-wave Ansatz with *two* momenta $k_{1,2}$ at the same energy; both lie on the bulk dispersion and the open boundaries quantise them [4, 5]:

$$\cos k_1 + \cos k_2 = -\frac{\mu\,t}{t^2-\Delta^2}, \qquad \frac{\sin^2\!\big[(L+1)k_\Sigma\big]}{\sin^2\!\big[(L+1)k_\Delta\big]} = \frac{1+(\Delta/t)^2\cot^2 k_\Delta}{1+(\Delta/t)^2\cot^2 k_\Sigma}, \tag{2.10}$$

with $k_{\Sigma,\Delta} = (k_1 \pm k_2)/2$. Real $k_{1,2}$ are bulk standing waves, complex $k_{1,2}$ the sub-gap edge states, and three finite-size lessons follow. (i) Since superconductivity puts $\pm\mu$ into the particle and hole blocks of (2.2), μ cannot act as a rigid energy shift: it enters the quantization itself, $k_n = k_n(\mu, t, \Delta)$, so the spacing is generically non-equidistant and every level drifts as μ is swept; the particle-in-a-box values $n\pi/(L+1)$ survive only at the level crossings and on the zero-energy lines (2.8) [5]. (ii) Edge modes are the bulk solutions that go missing: some real-k solutions vanish inside the window and re-enter with complex momentum. Eq. (2.4) is one, $k = i/\xi$ at $\mu < 0$ — purely evanescent, hence the monotonic sweet-spot splitting, whereas a generic $\mathrm{Re}\,k$ at $t \ne \Delta$ makes the two evanescent waves interfere. (iii) The boundary conditions exclude $k = 0, \pi$, so the gap closing at $|\mu| = 2t$ is unreachable at finite L: the spectrum only pinches to a finite-size gap.

3 Methods and results

All results are exact diagonalisations of the BdG matrix (2.2) at $t = \Delta = 1$ unless stated otherwise, obtained with the symmetric eigensolver `numpy.linalg.eigvalsh` in double precision. Quasiparticle energies are taken as $\mathrm{sort}(|E|)$ keeping one entry per $\pm E$ pair, rather than by filtering on the sign of E.

3.1 OBC spectrum and the two mid-gap levels

Figure 1 sweeps μ through both critical points at $L = 30$. A near-zero branch spans the whole topological window; beyond $|\mu| = 2t$ no sub-gap state remains. At $\mu = -t$ it sits at $E_0 = 1.397 \times 10^{-9}$, matching $2t(1-\frac{1}{4})(\frac{1}{2})^{30}$ from Eq. (2.5) to six significant figures; at $\mu = 0$ it is 2.5×10^{-16}, zero to double precision. The branch *is* the hybridised Majorana pair; the rounded merger at $|\mu| = 2t$ is the finite-size shadow of the bulk transition (see [Schewtschik, Part 1]), exact closing being forbidden at finite L by lesson (iii) of Sec. 2.5. Figure 2 resolves this state by state: the $L = 20$ spectrum is gapped at $\mu = -3t$ but carries two near-zero eigenvalues inside the gap at $\mu = -t$ — not two excitations but the $\pm E_0$ pair of a *single* level, the non-local fermion built from the two edge Majoranas.

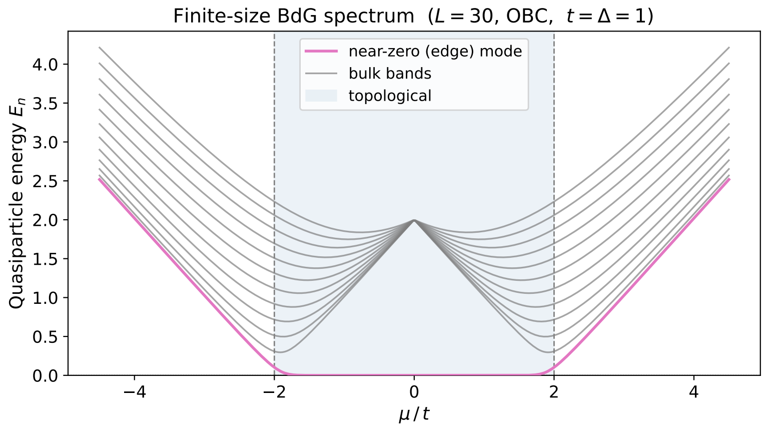

Figure 1: Lowest twelve quasiparticle levels of the $L = 30$ open chain vs μ: the near-zero edge branch spans the topological window and merges into the bulk continuum at $|\mu| = 2t$.

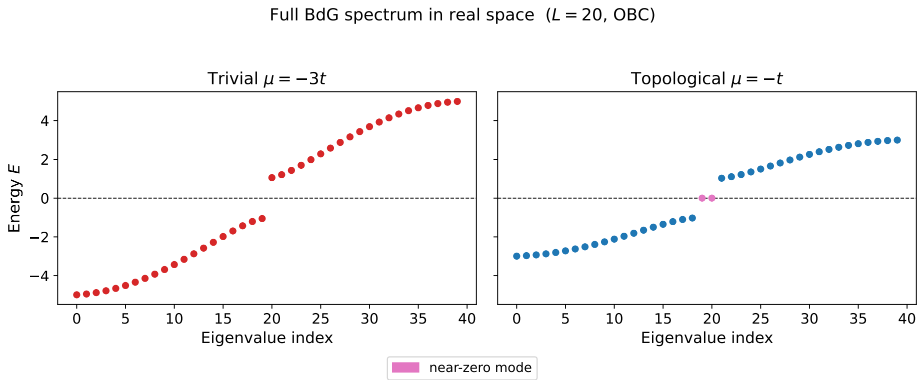

Figure 2: Full BdG spectrum of the $L = 20$ open chain: trivial phase (left, no sub-gap states) versus topological phase (right, a single $\pm E_0$ near-zero pair isolated inside the gap).

3.2 Exponential protection: E_0 versus L

Figure 3 shows $E_0(L)$ for $\mu/t = -0.5, -1, -1.5$ against Eq. (2.5) (dashed). The data are straight on a semilog scale with fitted slopes reproducing $\ln|\lambda|$, and the ratios $E_0/|\lambda|^L$ reproduce the prefactor $2t(1-\lambda^2)$, ruling out the bare estimate $2t|\lambda|^L$ (Table 1). The slope measures the coherence length, $\xi = 1/|\mathrm{slope}|$: 0.7 sites at $\mu = -0.5t$ but 3.4 at $\mu = -1.5t$, so protection weakens toward the phase boundary. Where the data leave the theory lines the splitting has fallen below machine precision (at $\mu = -0.5t$, $L = 40$ it should be 1.55×10^{-24}) and the measured values saturate in the shaded band.

3.3 Away from the sweet spot: Majorana lines and parity switches

Figure 4 tests Eq. (2.8) on one $L = 20$ chain at two pairings. At $\Delta = t$ the splitting is a smooth valley whose only exact zero is $\mu = 0$, where all Majorana lines coalesce. At $\Delta = 0.4t$ it instead oscillates beneath

Table 1: Fits to Fig. 3: slopes of $\ln E_0$ vs L over the even-L grid $6, 8, \ldots, 24$; ratios at $L = 20, 40, 100$ for $\mu/t = -0.5, -1, -1.5$ respectively (beyond those L the splitting has sunk below machine precision and the ratio is meaningless).

μ/t	fitted slope	$\ln|\lambda|$	$E_0/|\lambda|^L$	$2t(1-\lambda^2)$	$\xi = 1/|\mathrm{slope}|$
-0.5	-1.3866	-1.3863	1.8748	1.875	0.72
-1.0	-0.6932	-0.6931	1.5000	1.500	1.44
-1.5	-0.2907	-0.2877	0.8735	0.875	3.44

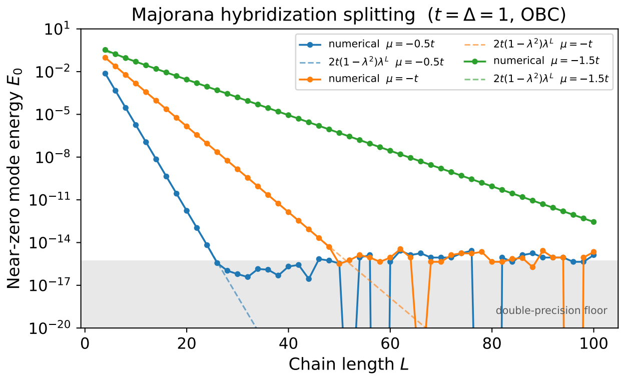

Figure 3: E_0 vs L with Eq. (2.5) dashed: slope $-1/\xi$, saturating in the shaded precision band.

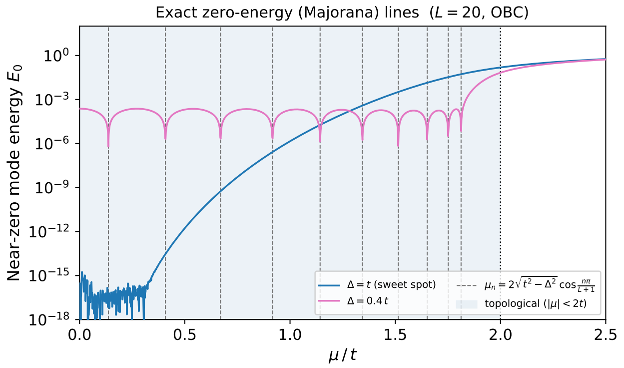

Figure 4: $E_0(\mu)$ at $L=20$: smooth at $\Delta=t$; oscillating at $\Delta=0.4t$ with exact zeros on the lines μ_n of Eq. (2.8) (dashed).

its envelope and dips to an exact zero at each of the ten predicted $\mu_n>0$ (dashed), with no fit parameter. Diagonalisation at $\mu=\mu_n$ gives E_0 between 3×10^{-18} and 3×10^{-16}, zero to double precision, against $E_0\approx2\times10^{-4}$ midway between lines — twelve orders of magnitude apart. Each μ_n is a level crossing at which the ground-state fermion parity switches. Since a real wire is never fine-tuned to $t=\Delta$, this oscillatory splitting — the lattice cousin of the Majorana oscillations of semiconductor nanowires [1] — is the generic finite-size fingerprint, and Fig. 3 its cleanest special case.

3.4 The sign-filter artefact

Extracting the "positive spectrum" naively as `evals[evals >= 0]` produces dense spikes over most of the topological window at large L (Fig. 5, left), while $|E|$ on the *same* eigenvalues is smooth (right). The spikes are a floating-point artefact, not physics [2]: the near-zero eigenvalue drops below machine precision once

$$E_0<\varepsilon_{\mathrm{mach}}\quad\Longleftrightarrow\quad|\mu|\lesssim2t\,\varepsilon_{\mathrm{mach}}^{1/L},\tag{3.1}$$

i.e. $|\mu|_{\mathrm{thresh}}\approx0.62$ at $L=30$ but ≈1.41 at $L=100$ for $\varepsilon_{\mathrm{mach}}=5\times10^{-16}$. Inside that window rounding can store the near-zero eigenvalue as a tiny *negative* number $\sim-10^{-16}$; the filter drops it and reports the lowest *bulk* level as "E_0" — at $\mu=-1.38t$, $L=100$ it returns 0.622 where $|E|$ gives 3.1×10^{-15}. Since the window *widens* with L, the spikes worsen for longer chains even though Eq. (2.5) says longer chains protect the mode better — the counter-intuitive observation that motivated this analysis. The lesson is to extract a symmetry-pinned quantity so that it survives ε-sized violations of that symmetry — here $|E|$ together with pair deduplication, which is the extraction used throughout this report.

4 On the AI collaboration

An LLM mined the block notes and reproduced the chiral-basis determinant argument of Sec. 2.4 from Ref. [5] in the convention of Eq. (2.1). Three corrections came out of that exchange: the measured ratios (Table 1) forced the $(1-\lambda^2)$ prefactor onto an initially bare $2t|\lambda|^L$; a physical reading of the large-L spikes was dropped once Eq. (2.5) was seen to predict the opposite trend; and an adversarial re-check caught a sign slip in Eq. (2.3) that had propagated into λ.

Compressing the material to the page limit proved the more instructive half. What to cut is a physics judgement as much as an editorial one — a step earns its place only if the argument cannot be reconstructed without it — and settling that forced an explicit view of what each section is for. Presentation is a technique

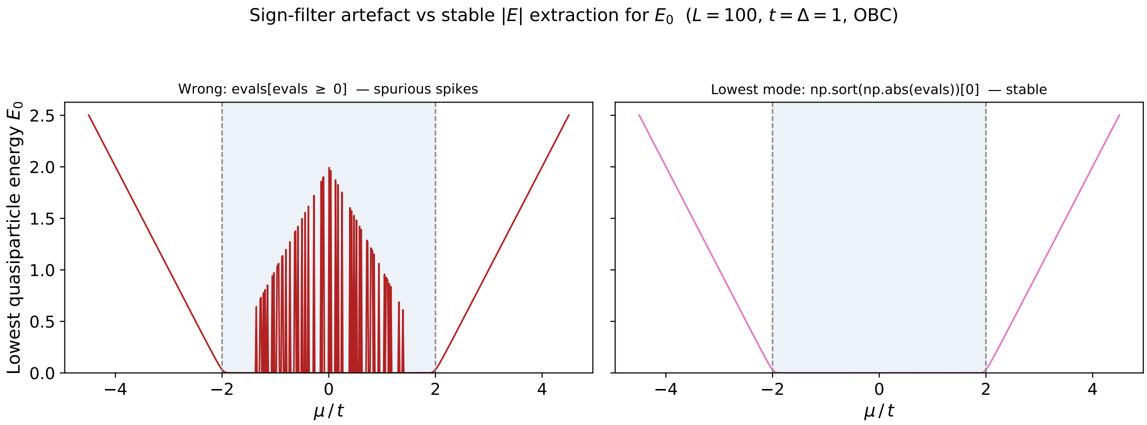

Figure 5: Same eigenvalues, two extractions $(L = 100)$: the sign filter (left) fabricates bulk-scale spikes over the window of Eq. (3.1); $|E|$ (right) resolves the near-zero mode.

in its own right, to be practised like the physics: redrafting against the model sharpened the register, showing where a hedge is honest and where it is padding, and how a scientific sentence is built to carry one claim at a time. Every number was re-verified against the code; the draft is to be revised by the author.

References

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